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Accelerating symmetry-breaking charge separation in a perylenediimide trimer through a vibronically coherent dimer intermediate

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Understanding the photophysics and photochemistry of molecular π-stacked chromophores is important for utilizing them as functional photonic materials. However, these investigations have been mostly limited to covalent molecular dimers, which can only approximate the electronic and vibronic interactions present in the higher oligomers typical of functional organic materials. Here we show that a comparison of the excited-state dynamics of a covalent slip-stacked perylenediimide dimer (2) and trimer (3) provides fundamental insights into electronic state mixing and symmetry-breaking charge separation (SB-CS) beyond the dimer limit. We find that coherent vibronic coupling to high-frequency modes facilitates ultrafast state mixing between the Frenkel exciton (FE) and charge-transfer (CT) states. Subsequently, solvent fluctuations and interchromophore low-frequency vibrations promote CT character in the coherent FE/CT mixed state. The coherent FE/CT mixed state persists in 2, but, in 3, low-frequency vibronic coupling collapses the coherence, resulting in ultrafast SB-CS between the distal perylenediimide units.

Photoinduced symmetry-breaking charge separation (SB-CS)[1–3] occurs in systems ranging from the chlorophyll special pair in photosynthetic reaction centre proteins[4,5] to organic photovoltaics[6,7]. It is thus important to understand the fundamental mechanism of SB-CS as well as other competing relaxation pathways in chromophore oligomers, such as excimer formation and singlet fission (SF), to design functional organic materials for photonic applications. The 'oligomer approach' employs molecular π stacks that mimic these systems using a small number of constituent units and has been successful for understanding structure–property relationships in discrete, well-defined constructs[8–11]. Among the various chromophores studied with this approach, perylenediimide (PDI) has been one of the most actively investigated because of its robustness, tunable absorption spectrum range, appreciable extinction coefficients[12] and propensity for strong π–π interactions.

In recent years, the molecular exciton theory developed by Kasha and Davydov[13,14] has been advanced by Spano and colleagues[15,16] to emphasize the vital role of short-range charge-transfer (CT)-mediated coupling in molecular aggregates[17–19]. This short-range coupling is very sensitive to the molecular packing geometry, such that sub-ångstrom displacements along the molecular long or short axis can impact the coupling strength. Thus, various photophysical pathways[20] such as excimer formation[10,21–28], SB-CS[29–32] or even SF[33–36] may be promoted or suppressed depending on the molecular packing geometry, along with the relative energies of the Frenkel excitons (FEs), CT states and correlated triplet pair states.

Accordingly, as shown in Fig. 1, we allocate the photoinduced dynamics of PDI dimer stacks into three categories in terms of the long-range Coulombic and short-range CT coupling interactions (Fig. 1a,b): (1) when the Coulombic coupling is small (longer interchromophore distance), the chromophores relax to the SB-CS state by an incoherent charge separation process[32]; (2) when the Coulombic coupling is strong (shorter interchromophore distance), structural relaxation can lead to the excimer state[26,37,38], which can act as an energy trap; (3) when both coupling strengths cancel, the chromophore stack lies in the so-called 'null-type' regime, because the overall electronic coupling strength is approximately zero. In the latter case, vibronic coupling may lead to electronic state mixing between the FE and CT states[36,39,40], although a null-type interchromophoric geometry is not a necessary condition for state mixing when the vibronic coupling is strong. In Fig. 1c we depict these cases in the coupling plane, including previous works on π-stacked PDIs. We assume that sufficient solvation takes place and exclude the SF channel and linear PDI assemblies for simplicity. Under these restrictions, there are four dimers composed of the same PDI unit but having different geometries compared to this work. One dimer possesses an interchromophore distance of 3.5 Å and forms an excimer state upon excitation[26], whereas the other three dimers have distances of 6.4–10.8 Å, with little orbital overlap, and undergo SB-CS[32]. It should be noted that we explicitly neglect through-bond interactions in this comparison, which should be negligible when the number of bonds in the linker is large, as is the case in this study, although they can result in SB-CS at intermediate interchromophore distances, where direct coupling through the chromophore π orbitals is weak[32].

In this Article we report the synthesis and characterization of a null-type PDI dimer (**2**) and trimer (**3**) to explore the role of vibronic coupling in coherent SB-CS. Using complementary electronic transient absorption (TA) and time-resolved infrared and vibrational coherence (VC) spectroscopies, in dimer **2** we observe rapid mixed-state formation between the adjacent PDI units in 200 fs by vibronic coupling along the high-frequency coordinates, whereas incoherent SB-CS takes place in PDI trimer **3** via electron or hole transfer from the initial coherent FE/CT mixed state to the additional distal PDI within 2–3 ps. This incoherent SB-CS is accelerated by the interchromophore low-frequency vibrations that

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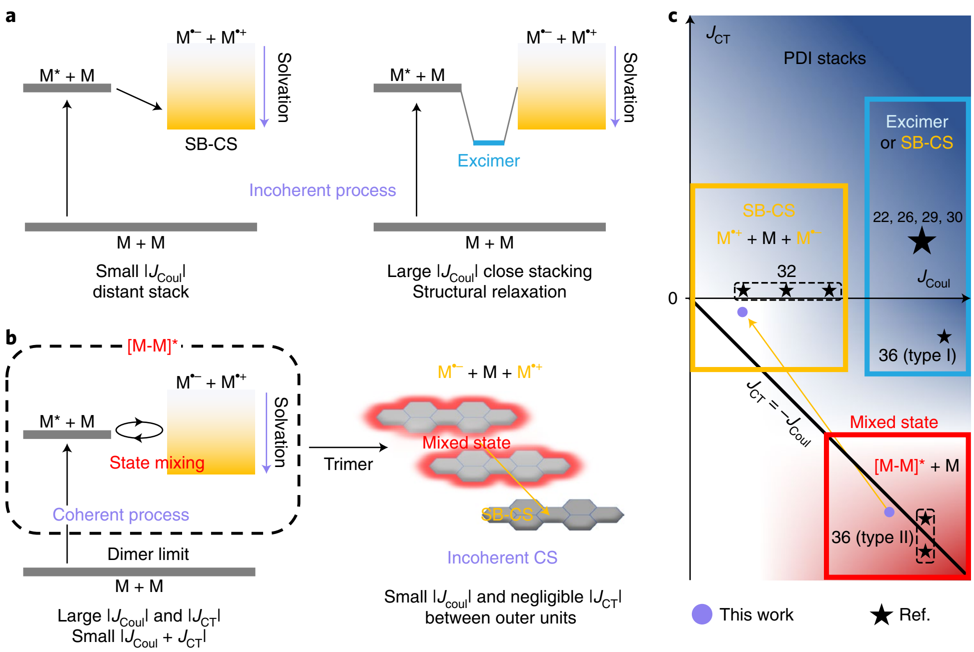

Fig. 1 | Representative photoinduced dynamics of PDI stacks. a, Typical SB-CS process (left) and excimer formation (right). **b**, Mixed FE/CT state formation, which further evolves to the SB-CS state between the outer PDI units in the trimer case. **c**, Previous studies and this work are plotted in the coupling strength plane along the long-range Coulombic and short-range CT coupling axes. This plot roughly considers the overall exciton coupling and geometry, except for compounds in ref. [36].

collapse the mixed state. Because this ultrafast SB-CS process in **3** is three orders of magnitude faster than in a dimer of similar chromophore separation[32], our findings provide fundamental insights into the efficient SB-CS enabled by the intermediate mixed electronic state. Additionally, we demonstrate that interactions between a coherently mixed dimer electronic state and a third PDI chromophore lead to rapid dephasing of the PDI dimer mixed state, which can have considerable implications for utilizing vibronic coherence in solid-state applications.

Results and discussion

Synthesis and molecular structures. The molecular structures of **1**, **2** and **3** are shown in Fig. 2a, with **2** and **3** synthesized by condensing the corresponding 1,6,7,12-tetrakis(4′-*tert*-butylphenoxy)perylene(3,4:9,10)bis(dicarboxydianhydride) or the corresponding monoimide-monoanhydride with the xanthene spacer (Supplementary Fig. 1). The PDI units are slipped by 4.3 Å along their long molecular axes and stacked with an average distance of 3.5 Å in both **2** and **3** (based on density functional theory (DFT) optimization; Supplementary Section 4). The π–π stacking distances of **2** and **3** are 3.5 Å and 6.9 Å, respectively, and the centre-to-centre PDI–PDI distance is 5.8 Å in **2** and the centre-to-centre distance between the two outer PDIs in **3** is 11.7 Å. Multi-dimensional NMR spectra (Supplementary Figs. 43–55) show the interchromophore proton correlations, which further support the stacked structures of **2** and **3**. Detailed analyses are provided in Supplementary Section 6.

Steady-state spectroscopy and electronic coupling strength. The UV–vis absorption spectra of **1**, **2** and **3** in THF are shown in Fig. 2b. The absorption maxima of **1**, **2** and **3** appear at 569, 581 and 585 nm, respectively. The redshifted 0–0 bands in **2** and **3** are consistent with the increased Coulombic coupling in J-type aggregates. Compared to **1**, both **2** and **3** show a slight increase in their I^{0-1}/I^{0-0} vibronic band ratios, which comes from weak H-type electronic interactions between PDI units. These increased ratios in **2** and **3** are small compared to those in cofacial H-type dimers showing inversion of the vibronic band ratio, which have competitive relaxation pathways between SB-CS and excimer formation[30]. The vibronic band ratio reports on the electronic coupling strength (J), which is described as the sum of long-range Coulombic (J_{Coul}) and short-range CT couplings (J_{CT}). Using the absorption spectra of **2** and **3**, and following the method developed in ref. [41], the calculated values of J are 15 and 46 cm^{-1} for **2** and **3**, respectively (Supplementary Fig. 3). Although estimating the couplings (J_{Coul} and J_{CT}) in the perturbative limit may be inaccurate when FE and CT states are not well separated[18], the cancellation of these values is also evident in **2** (Supplementary Section 2). This results from the presence of considerable opposite short-range CT coupling due to the substantial highest occupied molecular orbital–highest occupied molecular orbital (HOMO–HOMO) (lowest unoccupied molecular orbital–lowest unoccupied molecular orbital (LUMO–LUMO)) orbital overlap of **2**, which splits the MO levels (Δ_{HOMOs} of 1,097 cm^{-1} and Δ_{LUMOs} of 1,756 cm^{-1}; Supplementary Fig. 4), in keeping with differential pulse voltammetry measurements (Fig. 2c and Supplementary Fig. 42). Owing to the small electronic coupling strengths ($|J| < 100$ cm^{-1}) of **2** and **3**, they are classified as null-type aggregates[42,43]. As a result, the fluorescence spectra of **2** and **3** show monomeric shapes rather than the broadened excimer emission observed in cofacial PDI oligomers[10,30]. The substantial quenching of the fluorescence quantum yields observed for **2** and **3** in THF (both 0.015 ± 0.005) relative to **1** (0.90 ± 0.01) indicates the presence of rapid non-radiative deactivation processes. However, in toluene (Tol), the quantum yields of **2** (0.35 ± 0.01) and **3** (0.39 ± 0.01) remain at an intermediate level, indicating that these non-radiative processes depend on the solvent polarity (Supplementary Table 1). Additional steady-state data are shown in Supplementary Fig. 2.

TA spectroscopy. We performed TA spectroscopy to understand the general excited-state dynamics in three different temporal ranges. The TA spectra of **2** and **3** in THF, acquired on a narrowband femtosecond-excitation TA instrument (instrument response function (IRF) = 350 fs), are shown in Fig. 3a,b.

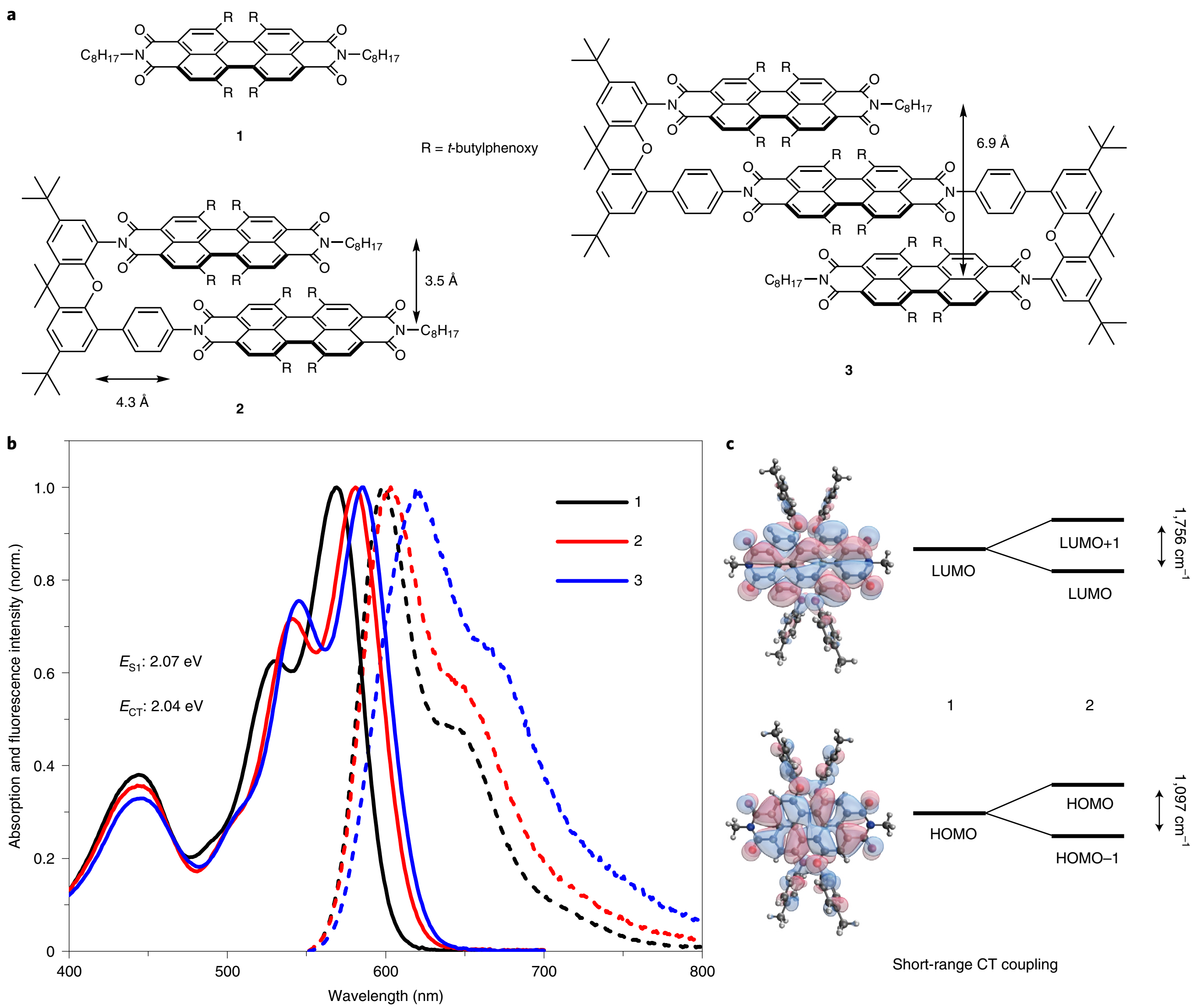

Fig. 2 | Molecular structures and steady-state results. **a**, Molecular structures of the monomer (**1**), dimer (**2**) and trimer (**3**), where R is a 4′-*tert*-butylphenoxy group and where rotation angles between neighbouring chromophores are ~15°. **b**, Steady-state absorption and fluorescence spectra of **1**, **2** and **3** in THF. **c**, Energy-level splitting between HOMOs and LUMOs in **2** due to the short-range CT coupling. E_{S1} and E_{CT} of **2** are denoted in **b**.

The evolution-associated spectra (EAS) were obtained by global fitting to an A → B → C → ground state model, where the last lifetime was determined from the nanosecond TA data (Supplementary Figs. 10 and 16). The sub-picosecond regime was studied using broadband-excitation TA measurements (IRF = 20 fs), with the 20-fs output of a noncollinear optical parametric amplifier used as the pump. Additional narrowband and broadband TA data are shown in Supplementary Figs. 5–18 and 21–27, respectively. Following broadband photoexcitation, the initial TA spectra of **2** and **3** in THF show dominant contributions from the FE state (710 and 950 nm), ground-state bleach (GSB, 581 nm) and a slight contribution from the CT state (636 and 793 nm) (Fig. 3c and Supplementary Fig. 24), where the FE band was assigned by matching the EAS spectra of **1** and the CT band observed in previous studies[32,44]. The CT character of this mixed state arises in 200 ± 20 fs and results in an intermediate state (Fig. 3d). Note that both the initial and intermediate states contribute to state A in narrowband-excitation TA measurements; for this reason, we label the broadband EAS as states A1 and A2, respectively. The dynamics and spectra of **2** and **3** in Tol are also similar at this timescale, having 200-fs components with increasing CT character, and the fast dynamics appear regardless of the solvent polarity (Supplementary Fig. 26). Because Frenkel exciton delocalization across more than three units is known in PDI stacks[25,42], the similarity of the mixed-state formation times and spectra for the dimer and trimer suggests that **3** rapidly localizes into a mixed state of two adjacent units, as the outer units have negligible CT coupling. Subsequently, the intermediate states further evolve to state B (Fig. 3a,b) with more CT and less FE character, with time constants of 2–3 ps, whereas **2** still has more FE character than **3** (Supplementary Fig. 27). Notably, the lifetimes of state B are different in **2** (2.30 ± 0.06 ns) and **3** (3.8 ± 0.4 ns), indicating that adding a third chromophore alters the nature of this state, even though the TA spectra remain similar. State B decays to triplet excitons (state C) that live for ~200 μs (Supplementary Fig. 10). In Tol, after state B is generated in 6–7 ps, we observe ~100-ps components for **2** and **3**, which we assign to slow structural relaxation (Supplementary Fig. 7) followed by decay in 8.6 ± 0.6 and 8.9 ± 0.6 ns, respectively. The SB-CS and charge recombination kinetics of **3** provide evidence

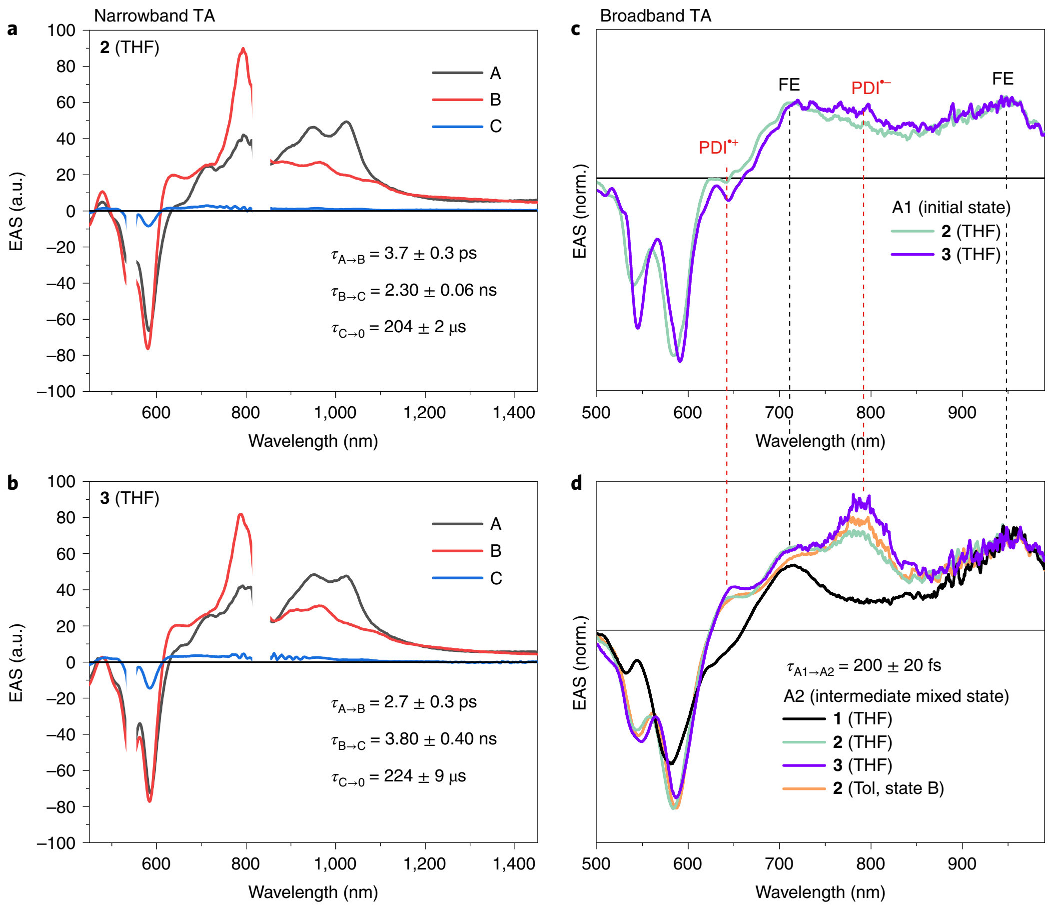

Fig. 3 | TA spectra (narrow and broadband excitation) and analysis. a,**b**, EAS from narrowband TA measurements for **2** (**a**) and **3** (**b**). **c**,**d**, EAS spectra before reaching state B (in narrowband measurements) obtained by broadband TA measurements of the initial state (**c**) and the intermediate mixed state (**d**). Dashed lines indicate FE (black) and charged-species (red) bands. Note that the A2 states of **2** and **3** show both FE and CT character.

for localization of the charges on the distal PDI units. The mixed-state lifetimes of **2** and **3** in Tol, which represent the 'dimer limit', are the same, whereas, in THF, full SB-CS does not occur in **2**, but proceeds rapidly from the initially formed FE/CT mixed state in **3**. Moreover, the charge recombination lifetime of the radical ion pair of **3** in THF is 65% longer than the FE/CT mixed-state lifetime in **2**, which is consistent with the reduced electronic coupling expected for radical ions localized on the distal PDIs (Supplementary Table 2). Although the TA results imply the formation of a mixed electronic state with both FE and CT character, we will employ time-resolved vibrational spectroscopy to provide additional evidence for this assignment.

Transient infrared spectroscopy. To probe the FE/CT mixed state with higher spectral resolution, we performed femtosecond infrared (fsIR) absorption (*A*) measurements (Fig. 4 and Supplementary Figs. 28–30) probing from 1,100 to 1,750 cm^{-1}, where the signature peaks from the FE and PDI radical anion/cation appear[32]. The fsIR spectra of **1** and charged species ($\mathbf{1}^{\bullet-}$ and $\mathbf{1}^{\bullet+}$) are displayed at the bottom, and the calculated fsIR spectra of the charged species are in good agreement with the experimental results (Supplementary Fig. 31). Note that the instrument response of the fsIR instrument is ~500 fs, limiting the observation of FE state evolution to the mixed FE/CT state in **2** and **3**. We explored five spectral regions to demonstrate that **2** forms a mixed FE/CT state, whereas **3** rapidly forms the fully separated SB-CS state. Specifically, in region I (1,135–1,212 cm^{-1}), containing FE and radical ion features, **2** exhibits strong and broad signals covering all these features, whereas the spectra of **3** show similar characteristics to those of the pure charge-separated species. Region III (1,442–1,500 cm^{-1}) has distinct features from FE (1,459 cm^{-1}) and charged species (1,482 cm^{-1}); whereas **2** displays features from both states, **3** shows a flat absorption profile and low signal at the FE region, and this weak signal diminishes rapidly. Additional analyses on the other regions (II, IV and V) are provided in Supplementary Section 3. The fsIR spectroscopy gives more detailed information on the dynamic state composition and the nature of the state mixing than electronic TA measurements, suggesting that states B in **2** and **3** from the electronic TA spectra discussed above are indeed different. The fsIR spectra for **2** indicate the coexistence of the FE and CT features, whereas, for **3**, the spectral features correspond to the fully separated SB-CS state.

Coherent wavepacket dynamics. To obtain fundamental insights into the VC in mixed-state formation and solvent relaxation, we carried out broadband TA measurements using an 11-fs compressed pump pulse, the bandwidth of which covers the steady-state absorption spectra of **1**, **2** and **3** (Supplementary Figs. 19 and 20) and is polarized parallel to the probe pulse. Fourier transform (FT) power maps (Fig. 5a,b) were obtained by Fourier-transforming the residual signals after subtracting the population dynamics from the TA signals (Supplementary Fig. 33). We can differentiate the VCs from the ground and excited states because the former predominantly lie in the 580–600-nm GSB region and the latter in the 630–680-nm region, which includes contributions from the PDI SE/ESA, FE and radical cation bands. Note that the PDI radical anion absorption band exists at 800 nm and is weak at 630–680 nm (ref. [32]), and thus does not contribute greatly. Also, detection of the

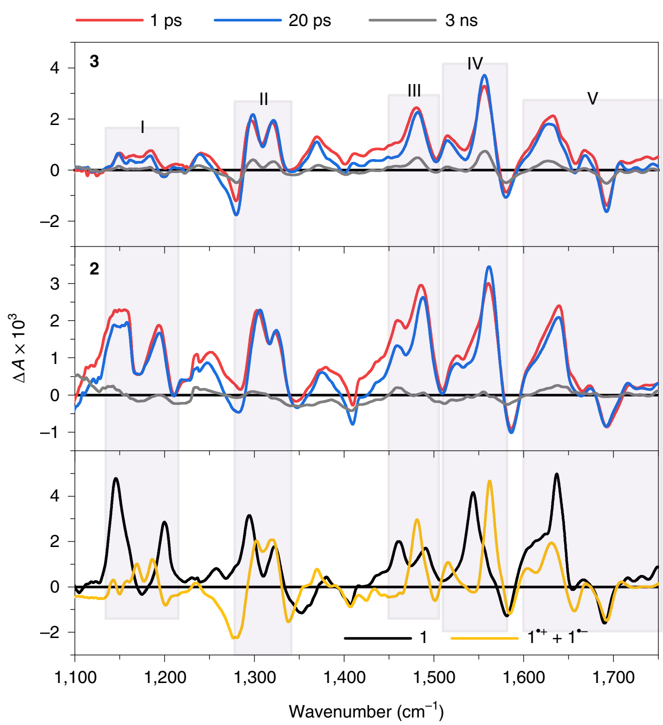

Fig. 4 | fsIR spectra of 1, 2 and 3 in CD_2Cl_2. The fsIR spectra of **1** at 10 ps and the charged species (bottom), and **2** (middle) and **3** (top) at three time delay points. Regions I–V are discussed in the main text. Excited-state IR-active modes effectively show the mixed (**2**) and SB-CS (**3**) states, as they are more sensitive to the electron density of chromophores than electronic transitions, which have broad signals and are congested with various signals.

radical anion region near 800 nm proved challenging due to temporal walk-off. This classification is also well reflected in the FT power spectra of **2** and **3** in THF, where most of the vibrational modes in the SE/ESA region are from the excited state (Supplementary Fig. 34). Surprisingly, numerous excited-state wavepackets were detected in **2** and **3**, including low-frequency (<1,000 cm^{-1}) and high-frequency (1,000–1,650 cm^{-1}) regions. The FT power spectra of **2** and **3** for the SE/ESA region are substantially different from those of **1**, as shown in Fig. 5c. The mode assignments of **2** and **3** are based on the spectrum of **1** and DFT calculations on the charged species (Supplementary Fig. 32).

Interestingly, although the residual signals of **1** and the solvents show typical dephasing behaviour, those of **2** and **3** have different oscillating patterns, especially in the low-frequency regions (Fig. 5d and Supplementary Fig. 36): using a temporally broadened pulse (~20 fs) to minimize coherent artefact signals near time zero (Supplementary Fig. 38), we found that, during formation of the mixed state (before 1 ps), **2** and **3** (THF) exhibit rapid dephasing of the oscillatory signals from the high-frequency modes compared to **1** (Supplementary Fig. 39). According to several previous works that have shown that vibronic coherence dephases on timescales intermediate to the purely vibrational and electronic limits[45–56], this observation indicates vibronic coupling along the high-frequency coordinates of **2** and **3**. Hence, the hastened dephasing of high-frequency modes in the dimeric and trimeric systems can be explained by vibronic coupling along these coordinates. A potential high-frequency vibration that may be coupled to the mixed-state formation is the core C=C stretching mode (~1,590 cm^{-1}), which is supported by its presence in the Fourier power spectra (Supplementary Fig. 37) and by a previous study of a CT reaction in a PDI compound[57]. Subsequently, multiple VCs assigned to the radical cation species (Supplementary Fig. 32) start to appear due to impulsive population transfer[58] or wavepacket evolution[59] coinciding with solvent relaxation after 1 ps. Most of the growing wavepackets are evident in the low-frequency region. The residual analysis also verifies the similarity of mixed states between **2** and **3** in Tol as their oscillatory signals are comparable (Fig. 5d). These dynamic oscillatory signals were further examined through the short-time Fourier transform (Fig. 5e) and the fast Fourier transform (FFT) filter method (Fig. 5f). Modes that shift or grow are assigned to PDI radical cation modes (260, 390, 465 and 660 cm^{-1}), and the others are assigned to FE modes (180 and 580 cm^{-1}) or solvent modes (Supplementary Fig. 35). Because the FE modes in the low-frequency region are present initially and dephase evenly across the molecular series, they are probably spectator modes orthogonal to the reaction coordinate[58]. In contrast, the low-frequency radical cation modes are presumably coupled to the photoinduced reaction along the solvent relaxation coordinate[60]. Three radical cation modes (260, 465 and 660 cm^{-1}), which become prominent after 1.5 ps in **2**, appear earlier than 1 ps in **3**. Afterwards, **3** exhibits different dynamics: the 260, 465 and 660 cm^{-1} radical cation bands diminish, and the other radical cation band appears at 390 cm^{-1}. These results suggest that the low-frequency radical cation modes of **3** are sequentially populated following the reaction from the FE to SB-CS state, where the mixed state acts as an intermediate between the two states.

Excited-state energy landscape. We can depict the excited-state potential energy surfaces of **2** and **3** along both generalized high-frequency PDI and low-frequency PDI and solvent coordinates (Fig. 6a). The rapid dephasing (<50 fs in THF) of high-frequency collective stretching modes at 1,100–1,650 cm^{-1} stems from coherent vibronic coupling that induces FE/CT state mixing by modulating the bond order between the π-stacked PDIs. As the mixed state is readily formed within 200 fs, regardless of solvent, vibronic coupling plays a dominant role along the relevant high-frequency coordinate. Although the solvent/low-frequency motions along the other coordinate can stabilize the CT state, increasing the CT character in the FE/CT mixed state and generating the multiple low-frequency VCs observed for the radical cation, strong vibronic coupling still results in FE dominance in the FE/CT mixed state, limiting the ability of the solvent to stabilize it.

Essentially the same FE/CT mixed state is formed initially in both **2** and **3**. However, the solvent is screened from the central PDI in **3**, because it is sandwiched between the two distal PDI molecules, each of which has one π surface that is fully solvated. Solvation of the distal PDI in **3** that is not involved in formation of the initial FE/CT mixed state results in lowering of the energy of the ion pair on the distal PDIs relative to that on the proximal ion pair, whose central ion cannot be as well solvated (Fig. 6b), resulting in the charges residing on the distal PDI molecules (Fig. 6b). SB-CS is also aided by interchromophore low-frequency vibrational modes that can substantially fluctuate the short-range CT coupling—for example, the out-of-plane interchromophore vibration (390 cm^{-1}) observed for the PDI radical cation species (Supplementary Fig. 40). This result thus emphasizes the role of the additional chromophore in collapsing the FE/CT mixed state, with the attendant formation of the desired SB-CS state comprising the distal PDI units.

Conclusions

We have synthesized a null-type, slip-stacked PDI dimer (**2**) and trimer (**3**) and explored the fundamental mechanism that drives ultrafast SB-CS in **3** using the electronic (TA) and vibrational (fsIR and VC) spectroscopies. Following photoexcitation, we find that a mixed FE/CT electronic excited state rapidly forms through coherent vibronic coupling primarily to a 1,590-cm^{-1} high-frequency mode. Solvent relaxation alters the balance between the FE and CT contributions, as indicated by growing VCs from the PDI

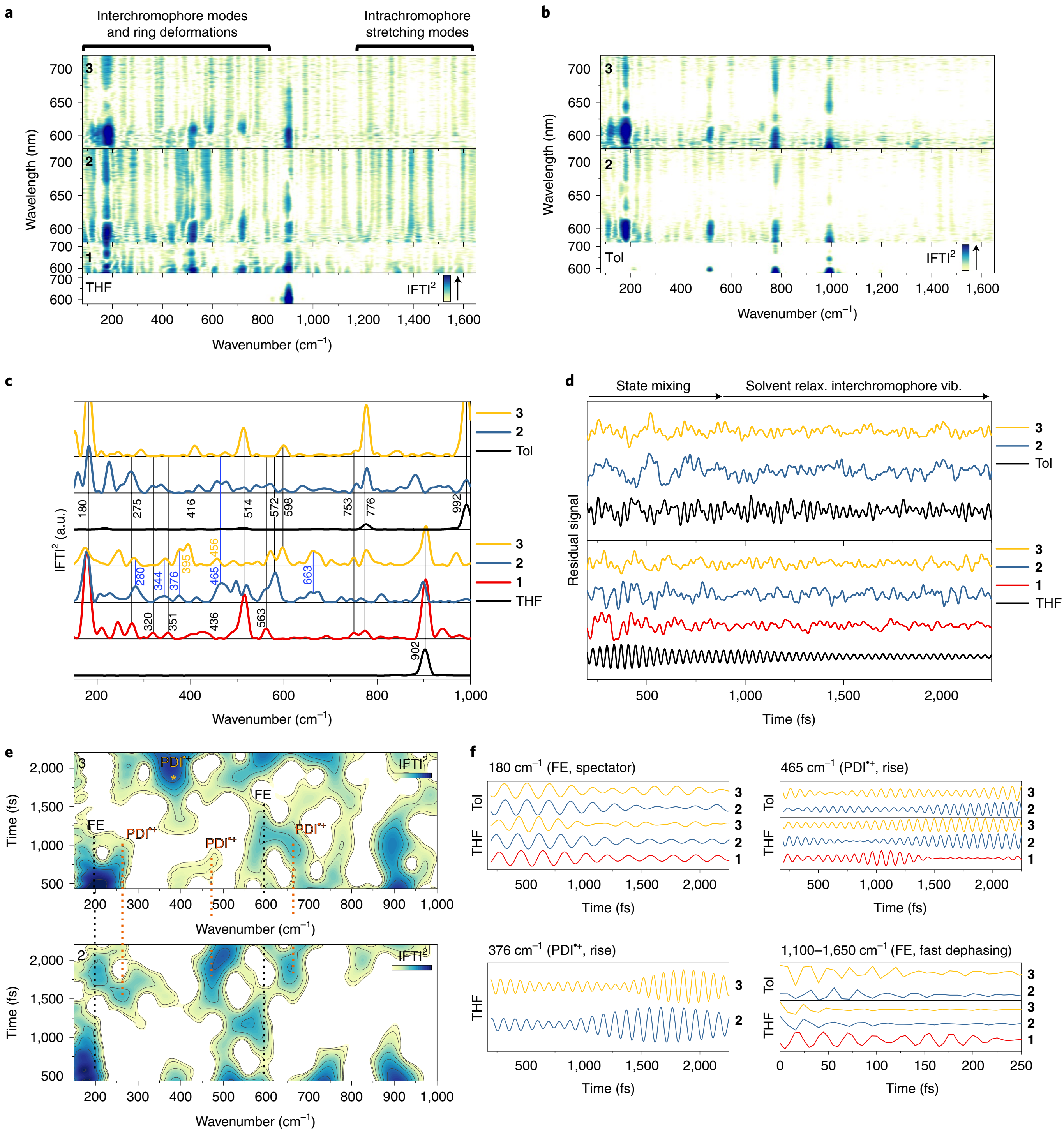

Fig. 5 | Vibrational coherence measurements and analysis. a,**b**, Fourier power maps of **1**, **2** and **3** in THF (**a**) and **2** and **3** in Tol (**b**). **c**, Corresponding FT power spectra in the low-frequency region for 630–680-nm SE and ESA signals, where vertical lines and coloured numbers denote FC active modes (black) and the evolution to the mixed state (blue), while further shifted modes in **3** are denoted with yellow numbers. **d**, Residual oscillatory signals for 630–680 nm (averaged) after subtraction of the population dynamics. **e**, Short-time FT analysis (Hanning sliding window) on **2** and **3** in THF, where dotted lines represent the FE (black) and mixed state (orange), which then further relaxes to the SB-CS state (yellow) in **3**. **f**, FFT filter analysis for several regions, as indicated. As Franck–Condon active modes are similar between chromophores, wavepacket evolution directly gives insights into how vibronic coherences correlate with the nature of the electronic state.

radical cation. Although **2** remains in the mixed state due to these vibronic and CT couplings, **3** reaches the SB-CS state by interacting with the adjacent chromophore, assisted by solvent relaxation and low-frequency interchromophore vibrational modes (for example, 390 cm^{-1}). The radical ion pair product probably resides on the distal PDI units where these couplings are no longer effective. Therefore, vibronic coupling ultimately accelerates SB-CS between the two outer PDI units, following collapse of the FE/CT

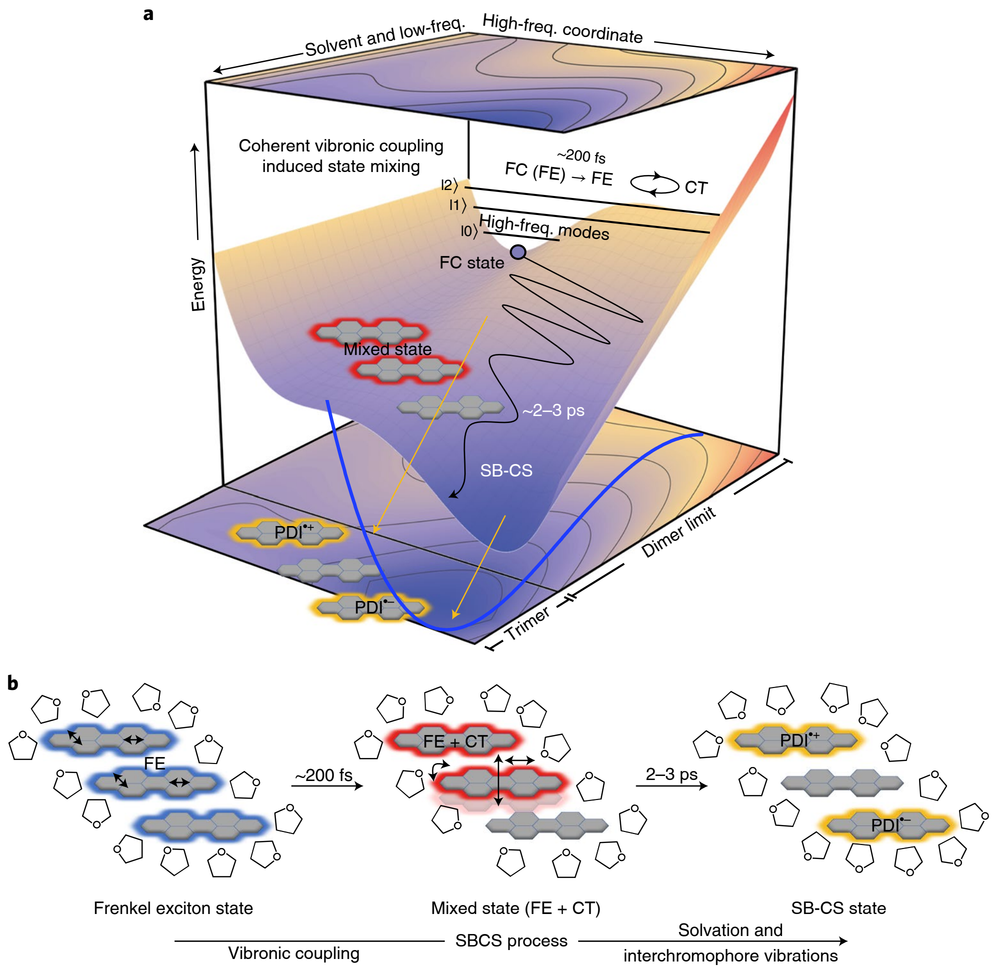

Fig. 6 | Potential energy surface. a, Potential energy surfaces involving the FE/CT mixed-state space (dimer limit) and SB-CS state in trimer **3** along the high-frequency and solvation/low-frequency coordinates where vibronic coupling facilitates state mixing followed by solvation/low-frequency collapse of the coherence and population of the SB-CS state. **b**, The extent of solvation increases in line with the photoinduced reaction in the trimer, which describes the lower energy level of the SB-CS state compared to that of the mixed state due to enhanced solvent–solute interactions.

mixed state involving two of the three PDI units. This work expands our fundamental understanding of the SB-CS process and provides design principles for efficient charge separation or SB-CS beyond the contact radical ion pair characteristic of dimers.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41557-022-00927-y.

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Methods

Synthesis. The preparation of **2** and **3** is described in detail in Supplementary Section 1.

Steady-state spectroscopy. The absorption spectra were measured on a Shimadzu UV-3600 UV/Vis/NIR spectrometer. The fluorescence spectra and quantum yield measurements were obtained using a Horiba Nanolog spectrofluorimeter (FL3-2iHR/iHR) equipped with an integrating sphere (Horiba Quanta-φ). The coupling strength was calculated following the method developed in ref. [41], which is described in detail in Supplementary Section 2.

Time-resolved spectroscopy. Narrowband fsTA measurements were performed on a Ti:sapphire laser system as described in detail previously[61]. Broadband fsTA and vibrational coherence measurements were performed using a Yb:KGW regenerative amplifier system with a noncollinear optical parametric amplifier (Spirit-NOPA, Light Conversion). The fsIR spectra were acquired following 400-nm excitation generated from the output of a commercial optical parametric amplifier (TOPAS-C, Light-Conversion), which has also been described previously[25]. Detailed descriptions and additional spectra are available in Supplementary Section 3.

Computations. The singlet ground-state geometries for **1**, **2** and **3** were optimized (Supplementary Fig. 41) using DFT at the level of M06-2X/6-31G(d,p) in QChem (version 5.1)[62]. We used time-dependent-DFT to compute the Coulombic coupling, as shown in Supplementary Section 2.

Data availability

Spectroscopic data and analysis, as well as the synthesis and characterization of **2** and **3**, are available in the Supplementary Information. Source data are provided with this paper.

Code availability

Codes for analysing the broadband TA data are available from M.R.W. upon request.

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Author contributions

C.L. synthesized and characterized the molecules, acquired the narrowband TA data, analysed the data and prepared the manuscript. T.K. acquired the broadband TA and fsIR data, analysed the data, performed the computational studies and prepared the manuscript. J.D.S. analysed the broadband TA data. M.R.W. and R.M.Y. designed the experiments, directed the investigations and prepared the manuscript with contributions from all the authors. All authors contributed to discussions.

Competing interests

The authors declare no competing interests.

Additional information

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